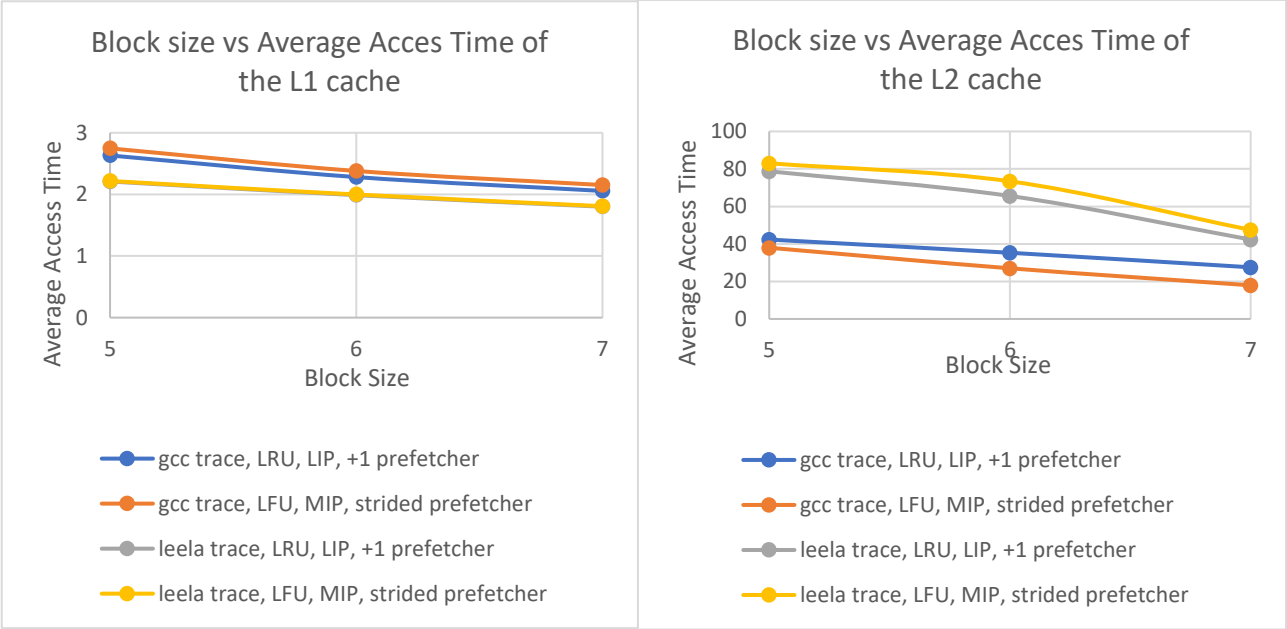
Félix Arnold

Experiments

We want to find the optimal cache configuration. To this end, I created a bash script running the cache for each trace and with each cache configuration possible.

We can observe that for all traces, the greater the block size *B*, the smaller is the average access time for both caches. On the following graph, we can see this plot for two different configurations of the *gcc* and the *leela* traces. The cache configuration was for L1: (15, B, 3) and for L2: (17, B, 8).



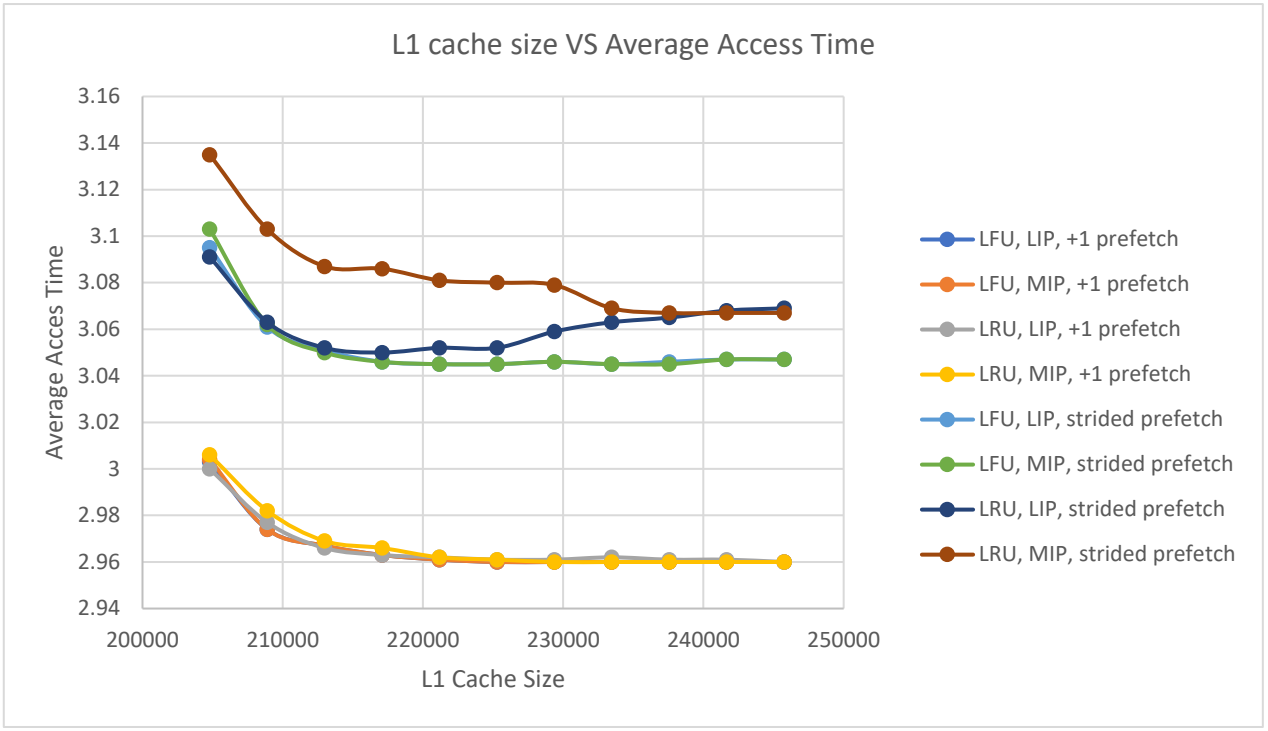
Thus, we can observe that a block size of 7 offer greater performances than a block size of 5. Moreover, the metadata storage is also smaller, with a size of 13568 bits for $B = 7$ and a size of 54272 bits for $B = 5$.

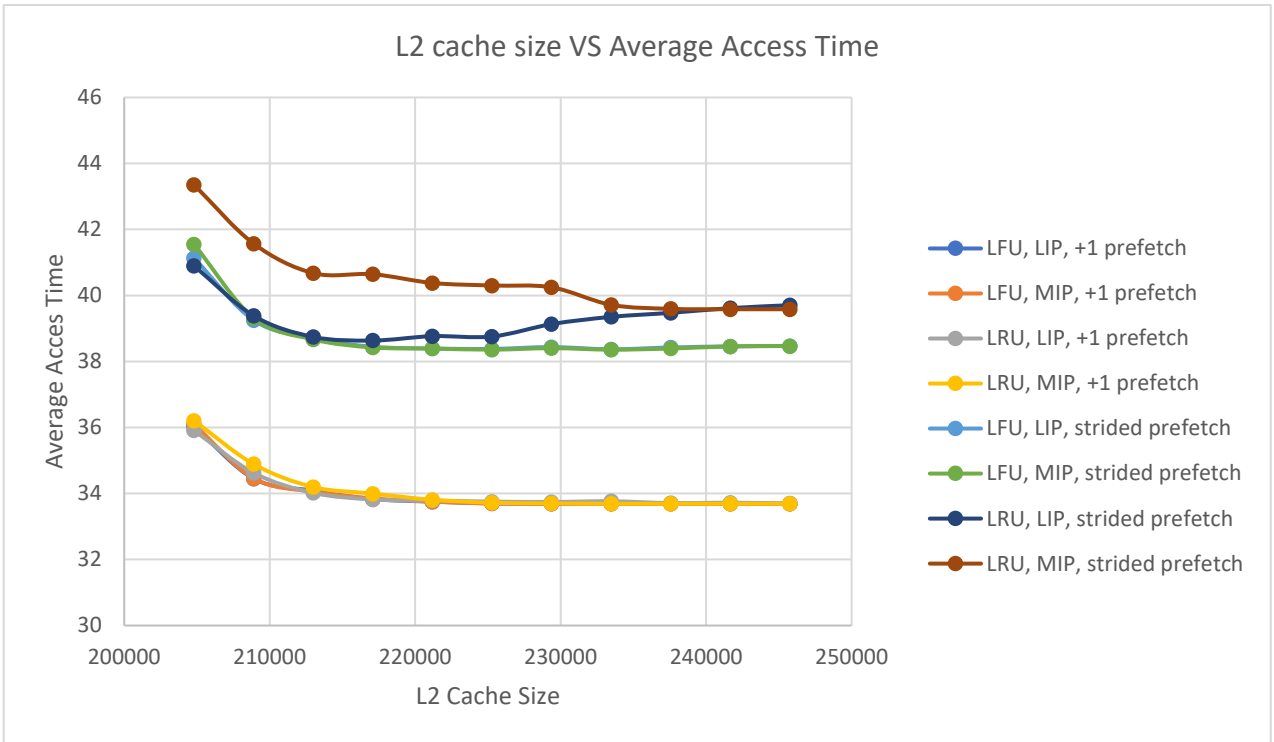
Then we can observe at the evolution of the cache metadata storage compared to the L1 and L2 cache size. Here are the output results for the *gcc* trace with L1: (15, 5, 1) and L2: (17, 5, S_2), with S varying between 2 and 12. We observe after a certain threshold, the average access time stop decreasing while the metadata storage size continues to increase.

Thus, the better trade of between is probably to select the value of S with the smaller metadata size while still having an average access time close to the limit.

As such, we can choose $S_2 = 3$. The value of AAT seems sufficiently low, and we have a cache storage size of only 212992 bits.

It seems also that the LFU insertion policy offers better value of average access time.





We can do the same by varying the L1 cache.